Complete Prometheus Installation Guide for Ubuntu

Prerequisites

```bash
# Update system packages
sudo apt update && sudo apt upgrade -y

# Install required dependencies
sudo apt install wget curl tar -y
```

Step 1: Create Prometheus User

```bash
# Create a system user for Prometheus (no login shell)
sudo useradd --no-create-home --shell /bin/false prometheus
```

Step 2: Create Directories

```bash
# Create directories for Prometheus configuration and data
sudo mkdir -p /etc/prometheus
sudo mkdir -p /var/lib/prometheus

# Set ownership
sudo chown prometheus:prometheus /etc/prometheus
sudo chown prometheus:prometheus /var/lib/prometheus
```

Step 3: Download and Install Prometheus

```bash
```

```bash
# Go to /tmp directory
cd /tmp

# Download Prometheus (check https://prometheus.io/download/ for latest version)
wget https://github.com/prometheus/prometheus/releases/download/v2.54.1/prometheus-2.54.1.linux-amd64.tar.gz

# Extract the archive
tar -xvzf prometheus-2.54.1.linux-amd64.tar.gz

# Navigate to extracted directory
cd prometheus-2.54.1.linux-amd64

# Copy binaries to /usr/local/bin
sudo cp prometheus /usr/local/bin/
sudo cp promtool /usr/local/bin/

# Set ownership for binaries
sudo chown prometheus:prometheus /usr/local/bin/prometheus
sudo chown prometheus:prometheus /usr/local/bin/promtool

# Copy console libraries and console files
sudo cp -r consoles /etc/prometheus
sudo cp -r console_libraries /etc/prometheus

# Set ownership
sudo chown -R prometheus:prometheus /etc/prometheus/consoles
sudo chown -R prometheus:prometheus /etc/prometheus/console_libraries
```

Step 4: Create Prometheus Configuration File

```bash
# Create the main configuration file
sudo nano /etc/prometheus/prometheus.yml
```

Add this basic configuration:

```yaml
```

```yaml
global:
  scrape_interval: 15s
  evaluation_interval: 15s

alerting:
  alertmanagers:
    - static_configs:
        - targets: []

rule_files: []

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']
```

Set correct ownership:

```bash
sudo chown prometheus:prometheus /etc/prometheus/prometheus.yml
```

Step 5: Create Systemd Service

```bash
# Create service file
sudo nano /etc/systemd/system/prometheus.service
```

Add this content:

```ini
```

```
[Unit]
Description=Prometheus
Wants=network-online.target
After=network-online.target

[Service]
User=prometheus
Group=prometheus
Type=simple
ExecStart=/usr/local/bin/prometheus \
  --config.file=/etc/prometheus/prometheus.yml \
  --storage.tsdb.path=/var/lib/prometheus/ \
  --web.console.templates=/etc/prometheus/consoles \
  --web.console.libraries=/etc/prometheus/console_libraries

Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

Step 6: Start Prometheus

```bash
# Reload systemd to apply new service
sudo systemctl daemon-reload

# Start Prometheus service
sudo systemctl start prometheus

# Check status
sudo systemctl status prometheus

# Enable Prometheus to start on boot
sudo systemctl enable prometheus
```

Step 7: Verify Installation

```bash
```

```bash
# Check if Prometheus is running
curl http://localhost:9090

# Check metrics endpoint
curl http://localhost:9090/metrics
```

Access Prometheus web UI at: `http://your-server-ip:9090`

Step 8: Install Node Exporter (System Metrics)

```bash
# Download Node Exporter
cd /tmp
wget https://github.com/prometheus/node_exporter/releases/download/v1.8.2/node_exporter-1.8.2.linux-amd64.tar.gz

# Extract
tar -xvzf node_exporter-1.8.2.linux-amd64.tar.gz

# Copy binary
sudo cp node_exporter-1.8.2.linux-amd64/node_exporter /usr/local/bin/

# Create user
sudo useradd --no-create-home --shell /bin/false node_exporter

# Set ownership
sudo chown node_exporter:node_exporter /usr/local/bin/node_exporter
```

Create Node Exporter service:

```bash
sudo nano /etc/systemd/system/node_exporter.service
```

Add:

```ini
```

```
[Unit]
Description=Node Exporter
Wants=network-online.target
After=network-online.target

[Service]
User=node_exporter
Group=node_exporter
Type=simple
ExecStart=/usr/local/bin/node_exporter

[Install]
WantedBy=multi-user.target
```

Start Node Exporter:

```bash
sudo systemctl daemon-reload
sudo systemctl start node_exporter
sudo systemctl enable node_exporter
sudo systemctl status node_exporter
```

Step 9: Configure Prometheus to Scrape Node Exporter

```bash
sudo nano /etc/prometheus/prometheus.yml
```

Add under `scrape_configs`:

```yaml
  - job_name: 'node_exporter'
    static_configs:
      - targets: ['localhost:9100']
```

Restart Prometheus:

```bash
sudo systemctl restart prometheus
```

Step 10: Install Alertmanager (Optional but Recommended)

```bash
cd /tmp
wget https://github.com/prometheus/alertmanager/releases/download/v0.27.0/alertmanager-0.27.0.linux-amd64.tar.gz

tar -xvzf alertmanager-0.27.0.linux-amd64.tar.gz

sudo cp alertmanager-0.27.0.linux-amd64/alertmanager /usr/local/bin/
sudo cp alertmanager-0.27.0.linux-amd64/amtool /usr/local/bin/

# Create user and directories
sudo useradd --no-create-home --shell /bin/false alertmanager
sudo mkdir /etc/alertmanager
sudo mkdir /var/lib/alertmanager

sudo chown alertmanager:alertmanager /usr/local/bin/alertmanager
sudo chown alertmanager:alertmanager /usr/local/bin/amtool
sudo chown -R alertmanager:alertmanager /etc/alertmanager
sudo chown -R alertmanager:alertmanager /var/lib/alertmanager
```

Create Alertmanager config:

```bash
sudo nano /etc/alertmanager/alertmanager.yml
```

Add basic config:

```yaml
global:
  resolve_timeout: 5m

route:
  receiver: 'default-receiver'
  group_wait: 10s
  group_interval: 10s
  repeat_interval: 1h

receivers:
  - name: 'default-receiver'
```

Create service:

```bash
sudo nano /etc/systemd/system/alertmanager.service
```

Add:

```ini
[Unit]
Description=Alertmanager
Wants=network-online.target
After=network-online.target

[Service]
User=alertmanager
Group=alertmanager
Type=simple
ExecStart=/usr/local/bin/alertmanager \
  --config.file=/etc/alertmanager/alertmanager.yml \
  --storage.path=/var/lib/alertmanager/

[Install]
WantedBy=multi-user.target
```

Start Alertmanager:

```bash
sudo systemctl daemon-reload
sudo systemctl start alertmanager
sudo systemctl enable alertmanager
```

Step 11: Configure Firewall (if UFW is enabled)

```bash
```

```
# Allow Prometheus port
sudo ufw allow 9090/tcp

# Allow Node Exporter port
sudo ufw allow 9100/tcp

# Allow Alertmanager port
sudo ufw allow 9093/tcp
```

Verification Commands

```bash
# Check all services
sudo systemctl status prometheus
sudo systemctl status node_exporter
sudo systemctl status alertmanager

# Check logs
sudo journalctl -u prometheus -f
sudo journalctl -u node_exporter -f
sudo journalctl -u alertmanager -f

# Test configuration
promtool check config /etc/prometheus/prometheus.yml
```

Access Points

- **Prometheus UI**: http://your-ip:9090
- **Node Exporter Metrics**: http://your-ip:9100/metrics
- **Alertmanager UI**: http://your-ip:9093

Common Issues & Solutions

Issue: Prometheus won't start

```bash
```

```bash
# Check configuration syntax
promtool check config /etc/prometheus/prometheus.yml

# Check permissions
ls -la /var/lib/prometheus
ls -la /etc/prometheus
```

Issue: Can't access web UI

```bash
# Check if service is running
sudo systemctl status prometheus

# Check if port is listening
sudo netstat -tulpn | grep 9090

# Check firewall
sudo ufw status
```

Issue: Targets showing as "down"

- Verify target services are running
- Check network connectivity
- Verify correct ports in prometheus.yml

Next Steps

1. Install Grafana for visualization
2. Set up alerting rules
3. Add more exporters for specific services (MySQL, Redis, etc.)
4. Configure retention policies
5. Set up remote storage (if needed)

Your Prometheus installation is now complete and ready to monitor your system!